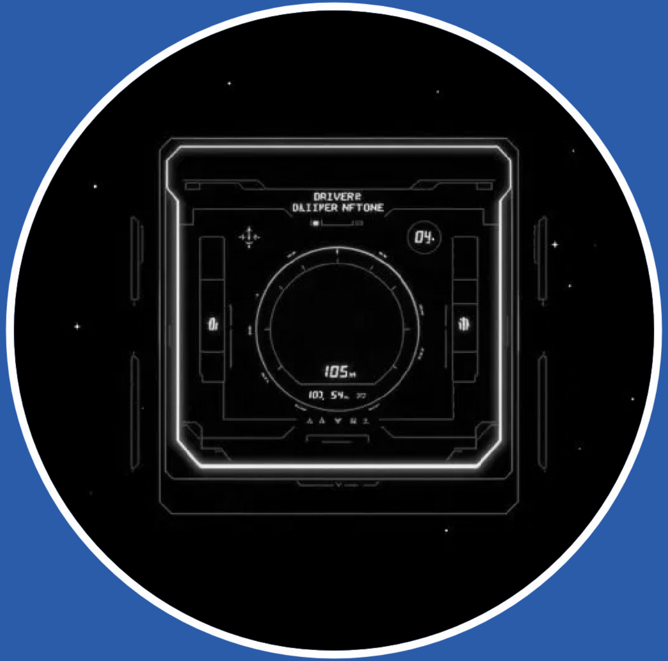
Applied AI System

AI Number Story Coach

Building a Reliable Applied AI Game System



Presenter: Amenuveve Tsetse

Python | OpenAI | RAG | Pytest

MOTIVATION

Evolution from Game to Product

Transforming a basic guessing game into a reliable AI-driven system.

THE ORIGINAL PROJECT

- Simple number guessing mechanics
- High / Low feedback loop
- Limited attempts win/loss state

THE STRATEGIC VISION

- Meaningful AI-driven strategy guidance
- Production-grade safety safeguards
- Focus on reliability and user trust

CORE RELIABILITY

- Input handling and validation logic
- Reasoning-based response generation
- Consistent fallback behaviors

SYSTEM UPGRADES

Key Enhancements

AI STORY COACH

Coaching after each guess with directional tips.

VISUAL FEEDBACK

Wordle-style progress tracking.

ADVANCED GAMEPLAY

Multiple difficulties and scoring.

LOCAL RETRIEVAL

Facts for numbers 1–500 with prioritization.

UI/UX EXCELLENCE

Theme persistence and tabbed navigation.

ROBUST CORE

Stronger validation and state reliability.

ARCHITECTURE

Engineering for Reliability

A modular pipeline designed to ensure safety, observability, and consistent performance.

MODULAR PIPELINE

- Input -> Validator -> State Engine
- AI Generator -> Safety Checker -> UI

RAG-ENHANCED MODE

- Local fact retrieval for numbers 1-500
- Optional document retrieval for context-aware coaching

SAFETY LAYERS

- Schema validation and directional correctness
- "No-answer-leak" protocols and fallback hints

OBSERVABILITY & FEEDBACK

- Comprehensive logging and agentic trace steps
- Automated tests and human review cycles

LIVE DEMO

Operational Reliability

Showcasing the balance of intelligent guidance and system control in a real-time environment.

- Robust Input Handling
- Agentic Transparency
- Persona Specialization

01 **Start Game:** Initialize on Medium mode to demonstrate challenge/guidance balance.

02 **Invalid Input:** Submit non-numeric guess to show toast alerts and attempt preservation.

03 **Repeated Guess:** Demonstrate logic blocking duplicate attempts without penalty.

04 **AI Coaching:** Submit valid guess to trigger the dynamic AI Story card generation.

05 **Agent Trace:** Expand the trace view to reveal intermediate reasoning and retrieval sources.

06 **Style Switch:** Toggle between Coach, Analyst, and Arcade personas in real-time.

07 **Persistence:** Show History and Stats tabs updating live with gameplay progression.

RELIABILITY

Guardrail Implementation

SAFETY PROTOCOLS

Automated checks for direction correctness, schema adherence, and answer leakage.

SYSTEM RESILIENCE

Fallback hint mechanisms ensure functionality during API outages or unsafe generations.

OBSERVABILITY

Comprehensive logging of system decisions and failure paths for continuous audit.

EVALUATION

Validation Results

METRIC CATEGORY	PERFORMANCE
Pytest Logic Coverage	**13 / 13 Passed**
AI Reliability Checks	**9 / 9 Passed**
Average Confidence Score	**0.70**
Directional Accuracy	**100%**
Schema Adherence	**100%**

**Results based on predefined scenarios and systematic reliability assessment via ai_evaluation.py.*

ADVANCED FEATURES

Advancing the AI Frontier

RAG ENHANCEMENT

Measurable impact from combining local number facts with external support documents for context-aware coaching.

AGENTIC WORKFLOW

Fully observable `agentTrace` steps providing deep insight into the AI's multi-step reasoning process.

SPECIALIZATION

Distinct personas (Coach, Analyst, Arcade) tailored to different user preferences and gameplay styles.

EVALUATION HARNESS

Dedicated `ai_evaluation.py` script for systematic reliability assessment across predefined scenarios.

REFLECTIONS

Key Learnings

- **Integration Details:** Reliability issues often stem from integration points rather than core logic.
- **Trust via Control:** Guardrails and fallback mechanisms are essential for building user trust in AI.
- **Architecture Speed:** A clear, modular design significantly accelerates debugging and iteration.

Responsible AI needs both intelligence and control.

FUTURE

Next Steps

EXPANSION

- Enrich the retrieval corpus with diverse number facts and quality metrics.

OBSERVABILITY

- Develop an in-app evaluation dashboard for real-time reliability monitoring.

ENGAGEMENT

- Expand to multiplayer and challenge modes while preserving safety checks.